

Print at 100% / Actual size, SINGLE-sided. Crease along the grey line, fold so both printed faces are outward, glue, then cut. This bar must measure exactly 80 mm.

[illegible]

ANALYSIS – BIG CUBES AND THE MAP

BIG CUBES

4x4 and 5x5

Reduce: build the centers → pair the 12 edges → solve it as 3x3. Row = 2 layers wide. 3Row = 2nd LAYER ONLY – never widen it.

last 2 edges

4x4 PLL parity

4x4 OLL parity – 1 edge pair flipped – half the time

5x5 edge parity

5x5 edge parity – 1 edge pair flipped – half the time – one move differs, 5x5 has no PLL parity

Fix parity during the last layer before OLL and PLL – both parity algorithms move corners.

NOTATION



Each letter tells a face clockwise, seen from outside the cube – so you turn your seat left, D and B look counter-clockwise – counter-clockwise = 2 half turn.

M = middle slice, turning the way L turns. x y z = the whole cube, turning the way R, U and F. Lowercase i = 1/2 that face plus the layer behind it, turning together.

SHORTCUTS

BEGINNER SHORTCUT – use it until you know Sune

Twist a yellow corner four turns at a time, turning ONLY the top face between corners.

yellow faces RIGHT R'D'R D x2

yellow faces FRONT R'D'R D x2

WORDS

algorithm = a fixed sequence of turns - trigger - a short algorithm your hands run as one unit - sexy move = the R U R' U' trigger - the R U R' U' trigger - parity = a case a 3x3 cannot have: edge pair = two edges glued into one, on a 4x4 or 5x5

RURU' sexy move RURU Sune R'FRF' sledgehammer gap = change grip

[illegible][illegible]

1 / FIRST SOLVE
 THE LADDER
 2 / 3VNO-LOOK-15 - algorithms - unlock five spaces and solve
 3 / 3VNO-LOOK-15 - unlock: there has to be a layer in exactly two
 4 / 3VNO-LOOK-15 - 15 - don't at all 21 moved cubes on separate
 After that first full TLL 222, solving planning look-ahead - in the pop.
WHY THERE ARE ONLY THREE
 A cube is good at a given set of cases you don't want by which it
 is in a never fails a combination work at this state. All of them are in
 when they are medium steps.
NOSE
 Every 4-LOOK you can solve is expanded from machine-generated data and checked against a cube simulator at build time. If one were
 to try to generate every possible permutation of the cube, it would take a point at 100% / 100% and would never fit the page.
 10mm

[illegible]

ANNEX – BIG CUBES AND THE MAP

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BIG CUBES

4x4 and 5x5
Reduce; build the centers → pair the 12 edges → solve it as a 3x3. Rw = 2 layers wide. 3Rw = 3 layers. 2R = 2nd LAYER ONLY → never widen it.

2nd edges

4x4 PLL parity

$\begin{matrix} R & U & R' & F' & R' & U \\ 2R^2 & U^2 & 2R^2 & U^2 \end{matrix}$

4x4 ULL parity = 1 edge pair flipped - half the time

Rw U2 x Rw U2 Rw U2 Rw' U2 Lw U2 Rw U2 Rw' U2 Rw' U2 Rw'
5x5 edge parity = 1 edge pair flipped - half the time - one move differs; 5x5 has no PLL parity

Rw U2 x Rw U2 Rw U2 **3Rw** U2 Lw U2 Rw U2 Rw' U2 Rw' U2 Rw'

5x5 edge parity = 1 edge pair flipped - half the time - one move differs; 5x5 has no PLL parity

NOTATION

Let every letter turn by face clockwise, seen from outside the cube – so from your seat L, D and B look counter-clockwise – counter-clockwise = ½ half turn.
M = middle slice, turning the way L turns. x y z = the whole cube, turning the way R, U and F turn. Lowercase i f = that face plus the layer behind it, turning together.

WORDS

algorithm = a fixed sequence of turns - trigger = a short algorithm you hands run as one unit - sexy move = the R U R' U' trigger - sledgehammer = the R' F R' U' trigger, RW is a case a 3x5 cannot have edge/pair = two edges glued into one, on a 4x4 or 5x5

RURU sexy move **RURU** Sexy **R'FRF'** sledgehammer case - glue chips

ANNEX – BIG CUBES AND THE MAP

BIG CUBES 4x4 and 5x5
Reduce; build the 6 centers → pair the 12 edges → solve it as a 3x3. Rw = 2 layers wide. 3Rw = 3 layers. 2R = 2nd LAYER ONLY – never invert!

last 2 edges
4x4 PLL parity $\begin{matrix} R & U & R' & F' & R' & U \\ 2R^2 & U^2 & 2R^2 & U^2 \end{matrix}$ slice out, 2 layers wide. dice back – both cubes $\begin{matrix} 2R^2 & U^2 & 2R^2 & U^2 \end{matrix}$ 2 edge pairs swapped - half the time

4x4 OLL parity 1 edge pair flipped - half the time
 $Rw\ U^2\ x\ Rw\ U^2\ U^2\ Rw\ U^2\ Rw'$ U^2 Differs w/o PLL parity $Rw\ U^2\ Rw'\ U^2\ Rw'$
5x5 edge parity 1 edge pair flipped - half the time - one move differs, 5x5 has no PLL parity
 $Rw\ U^2\ x\ Rw\ U^2\ Rw\ U^2\ 3Rw\ U^2\ Lw\ U^2\ Rw'$ slice out, 2 layers wide. dice back – both cubes $Rw\ U^2\ Rw'\ U^2\ Rw'$

NOTATION
Each letter turns that face clockwise,
seen from outside the cube – so from your seat L and B look counter-clockwise – counter-clockwise = 2 half turn.
M = middle slice, turning the way L turns. x y z = whole cube, turning the way R U L and F turn. Lowercase i f = that face plus the layer behind it, turning together.



WORDS
algorithm = a fixed sequence of turns - trigger = a short algorithm you hands runs as one unit - sexy move = the R U R' U' trigger -
downface = the F R F' trigger, parity = a case a 3x5 cannot have edge/pair = two edges glued into one, on a 4x4 or 5x5

algorithms
RURU' sexy move RURU'Sexy FRFPF' sledgehammer game change grip